Find relations between amplitude and phase of a plane wave with I/Q concept in radar terminology.

Solution

Let us assume that an incident ($\vec{\boldsymbol{E}}_i$) electric field of a monochromatic (time-harmonic) electromagnetic wave is excited from a radar antenna over a range r

$$\vec{\boldsymbol{E}}_i(\vec{\boldsymbol{r}},t) = E_0\cos(kr-\omega_0 t)\vec{r}$$

where *k* and ω_0 are wave number and source frequency, respectively. Using a complex notation concept, phasor-like form $E_i(\vec{\boldsymbol{r}})$ is formulated as

$$\vec{\boldsymbol{E}}_i(\vec{\boldsymbol{r}},t) = Re\{E_0 e^{jkr}e^{-j\omega_0 t}\vec{r}\} \quad \Longrightarrow \quad \vec{\boldsymbol{E}}_i(\vec{\boldsymbol{r}}) = E(\vec{\boldsymbol{r}})\vec{r} = E_0 e^{jkr}\vec{r}$$

$$E_i(\vec{\boldsymbol{r}}) = E_0 e^{jkr} = \underbrace{E_0\cos(kr)}_{I} + j\underbrace{E_0\sin(kr)}_{\mathrm{Q}} = \mathrm{I} + j\mathrm{Q}.$$

where I and Q are similar to I/Q concept (In-Phase and Quadrature) in radar terminology.

Now, let us show that E_0 and kr are the amplitude and the phase of the signal. Reconsidering the previous form of the $E_i(\vec{\boldsymbol{r}})$ again

$$E_i(\vec{\boldsymbol{r}}) = E_0 e^{jkr} = E_0\cos(kr) + jE_0\sin(kr)$$

where

$$Amplitude = \sqrt{(E_0\cos(kr))^2 + (E_0\sin(kr))^2} = \sqrt{E_0^2\underbrace{(\cos^2(kr)+\sin^2(kr))}_{=1}} = \sqrt{E_0^2} = E_0$$

$$Phase = arctan\left(\frac{E_0\sin(kr)}{E_0\cos(kr)}\right) = arctan\left(\frac{\sin(kr)}{\cos(kr)}\right) = arctan(\tan(kr)) = kr.$$

They can be related to I/Q concept as

$$Amplitude = E_0 = \sqrt{I^2+Q^2}$$

$$Phase = kr = arctan\left(\frac{Q}{I}\right).$$

Note: The I/Q signals are also known as active/reactive components.